Toward Complex Backgrounds: A Unified Difference-Aware Decoder for Binary Segmentation

Jiepan Li, *Graduate Student Member, IEEE*, Wei He, *Senior Member, IEEE*, Fangxiao Lu, *Student Member, IEEE*, and Hongyan Zhang, *Senior Member, IEEE*

***Abstract*—Binary segmentation is used to distinguish objects of interest from background, and is an active area of convolutional encoder-decoder network research. The current decoders are designed for specific objects based on the common backbones as the encoders, but cannot deal with complex backgrounds. Inspired by the way human eyes detect objects, we propose a new unified dual-branch decoder paradigm, termed the difference-aware decoder, to better explore the differences between foreground and background and to separate objects of interest in optical images. This decoder operates in two stages, leveraging multi-level features from the encoder. In the first stage, coarse detection of foreground objects is achieved by directly utilizing high-level semantic features, mimicking the initial rough observation of human vision. In the second stage, the decoder refines segmentation by exploring differences in low-level features, guided by the coarse map from the first stage. To enhance this process, we introduce two key innovations. First, a difference-aware prototype generation strategy leverages the guide map to extract foreground and background prototypes from high-level features, and calculates the similarity between these prototypes and corresponding representations in low-level feature spaces. Second, an overlapped window cross-level semantic guidance mechanism integrates high-level semantic information into low-level features through channel grouping and multi-scale aligned window pairs, guided by the similarities computed in the first strategy. Together, these innovations significantly enhance the DAD's ability to discern subtle differences, enabling precise foreground extraction and effectively addressing the challenges of complex and varied backgrounds. To verify the performance of the proposed difference-aware decoder, we choose three well known backbones including ResNet, Res2Net, PVT, and two binary segmentation tasks, *i.e.*, salient object detection, and camouflaged object detection, for comparative experiments. The results demonstrate that the difference-aware decoder can achieve higher accuracy than the other state-of-the-art binary segmentation methods for these tasks. The source code will be available on https://github.com/Henryjiepanli/DAD**

***Index Terms*—Dual-branch decoder, binary segmentation, salient object detection, camouflaged object detection, polyp segmentation.**

I. Introduction

BINARY segmentation aims to distinguish pixels belonging to the foreground from the background, a problem that has garnered substantial attention in the field of computer vision. As the scope of objects of interest expands, binary segmentation has become a crucial technique in various applications, including salient object detection [1], camouflaged object detection [2], and medical image segmentation [3]. Moreover, it plays a significant role in real-world tasks such as autonomous driving [4], and robot navigation [5].

In many image segmentation tasks, the relationship between foreground and background varies significantly, making binary segmentation a challenging problem. This task is often addressed using the widely adopted encoder-decoder architecture [6], where the encoder extracts multi-level features from the input image through a series of convolutional operations, and the decoder uses these multi-level features to generate segmentation results. Common encoders (backbones) include CNN-based models such as ResNet [7], and Res2Net [8], as well as Transformer-based architectures like Pyramid Vision Transformer (PVT) [9], Vision Transformer (ViT) [10], and Swin Transformer [11]. These multi-level features can be broadly classified into low-level and high-level features. Low-level features are rich in fine details such as texture and color but lack deep semantic understanding. On the other hand, high-level features capture strong semantic cues but tend to lose fine-grained details due to downsampling. Therefore, effectively integrating both low-level and high-level features in the decoder is crucial for improving the accuracy of binary segmentation, a challenge that has been the focus of substantial research in recent years.

The general decoding strategy of binary segmentation involves aggregating multi-level encoded features by enriching the multi-scale contextual information [2], [12], [13], capturing object-related attributes (such as frequency [14], texture [15], *etc.*), integrating the boundary-aware information [16], [17], [18], addressing uncertainty in predictions [19], [20], [21], or adopting a multi-stage refinement strategy [22], [23], to improve the final segmentation performance. Recently, some approaches [24], [25] have utilized conditional diffusion strategies to refine the boundaries of foreground objects iteratively, aiming to improve segmentation accuracy in challenging cases.

Despite these advances, decoding remains challenging in the presence of diverse and cluttered backgrounds. Most existing methods attempt to enhance foreground representations using coarse predictions or high-level activations from the encoder. However, these strategies often treat the coarse map merely as a spatial cue and fail to fully exploit the semantic abstraction available in high-level features. As a result, they struggle to clearly distinguish foreground from background in low-level representations, especially when the two are visually similar. This limitation motivates us to explore how high-level

Received 17 July 2025; revised 1 September 2025; accepted 18 September 2025. Date of publication 22 September 2025; date of current version 5 February 2026. This work was supported by the National Natural Science Foundation of China under Grant 42271370. This article was recommended by Associate Editor F. Lv. *(Corresponding author: Wei He.)*

Jiepan Li, Wei He, and Fangxiao Lu are with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430072, China (e-mail: jiepanli@whu.edu.cn; weihe1990@whu.edu.cn; fangxiaolu@whu.edu.cn).

Hongyan Zhang is with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430072, China, and also with the School of Computer Science, China University of Geosciences, Wuhan 430074, China (e-mail: zhanghongyan@whu.edu.cn).

Digital Object Identifier 10.1109/TCSVT.2025.3612574

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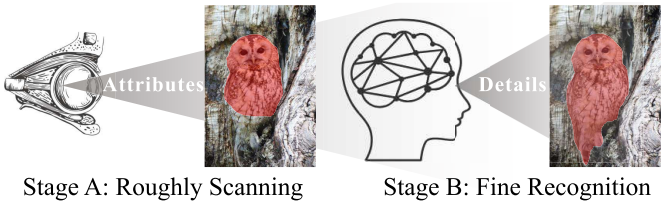


Fig. 1. The human visual process for object recognition.

semantics can be explicitly utilized to guide the refinement of low-level features more effectively.

To address this challenge, we propose a unified **dual-branch decoder paradigm named the difference-aware decoder (DAD) for the binary segmentation task**. This approach is inspired by the process of how the human visual system detects and recognizes objects [26]. As depicted in Fig. 1, the human visual system initiates scene analysis with a **Rough Scanning** phase, producing an initial segmentation map that identifies potential foreground regions. Guided by prior knowledge, this stage establishes a preliminary understanding of the object's category, location, and structure, encapsulating high-level semantic information. In the subsequent **Fine Recognition** phase, the brain focuses on subtle foreground-background distinctions, leveraging the prior semantic understanding to refine spatial details and enhance perceptual precision. Inspired by this mechanism, our framework employs the coarse map and high-level semantic features as priors to explicitly modulate low-level features. This semantic guidance disentangles ambiguous local representations into distinct foreground and background components, enhancing fine-grained separability and achieving precise boundary localization.

To embody this idea, the proposed DAD operationalizes semantic guidance through two complementary stages. The first stage generates a coarse guide map from high-level features, serving as a semantic prior that outlines the object's global location and structure. The second stage then refines low-level features under this guidance, with an emphasis on foreground–background discrepancies. Unlike conventional refinement pipelines that treat the coarse map merely as a spatial cue, DAD introduces two dedicated mechanisms to make semantic guidance both explicit and effective: (1) a **Difference-Aware Prototype Generation Module**, which extracts foreground and background prototypes from the guide map and high-level features, and leverages them to explicitly contrast low-level representations; and (2) an **Overlapped Window Cross-Level Semantic Guidance Mechanism**, which explicitly models semantic interactions between high-level features and multi-scale local windows of low-level representations. By operating on overlapped windows across scales, this mechanism injects rich contextual cues into adjacent regions, thereby enhancing the discriminability of ambiguous foreground–background boundaries. Through these designs, DAD transcends generic coarse-to-fine refinement by transforming high-level semantics into actionable priors, enabling robust disambiguation in cluttered scenes and precise recognition of challenging foregrounds.

The main contributions of this study are as follows:

1) We introduce a unified Difference-Aware Decoder (DAD) that emulates the human visual system's process of object detection and recognition for binary segmentation.
2) We propose the Difference-Aware Prototype Generation Module and the Overlapped Window Cross-Level Semantic Guidance Mechanism, which enhance the separability between foreground and background differences.
3) We perform extensive experiments on two binary segmentation tasks (*e.g.*, SOD, COD) using various backbone architectures to validate the effectiveness of the proposed DAD.

The paper is organized as follows: Section II reviews related work. Section III details the proposed difference-aware decoder. Experimental results and analysis are presented in Section IV, followed by ablation and parametric studies in Section V. Section VI concludes the paper.

II. Related Works

A. Decoders in Binary Segmentation

From the perspective of decoding mechanisms, approaches can be classified into several types: 1) **Multi-scale-context based approaches** [27], [28], [29], [30], [31]: These approaches develop multi-scale context modules to describe foreground objects in different scales and with diverse appearances; 2) **Mechanism-simulation based approaches** [2], [22], [32], [33], [34], [35]: These methods emulate some natural mechanisms by employing a multi-stage coarse-to-fine strategy to progressively refine segmentation results; 3) **Boundary-supervision based approaches** [16], [36], [37]: These approaches introduce boundary supervision as an auxiliary loss to achieve precise segmentation outcomes with more refined texture details; 4) **Frequency-domain integrated approaches** [14], [36], [38], [39]: These methods explore distinguishable foreground-related cues in the frequency domain, utilizing them as auxiliary information to better identify foreground objects in the spatial domain; 5) **Uncertainty-estimation based approaches** [19], [20], [21], [40], [41], [42], [43]: These approaches approximate uncertainty across different image regions, guiding the model to focus on areas that are difficult to identify, resulting in high-confidence segmentation. 6) **Diffusion-based approaches** [24], [25]: These methods utilize the denoising process of diffusion models to iteratively refine segmentation results.

1) Positioning Our Work: DAD redefines the coarse-to-fine paradigm by leveraging high-level semantic features as priors to guide low-level feature refinement, enabling explicit foreground-background disentanglement. Unlike existing methods that rely on coarse predictions as spatial masks, DAD utilizes semantic guidance to resolve low-level feature ambiguities, delivering superior segmentation performance in challenging scenes with visually similar regions.

B. Cross-Layer Interaction

Commonly employed strategies often rely on combining contextual modules with up-sampling operations to construct

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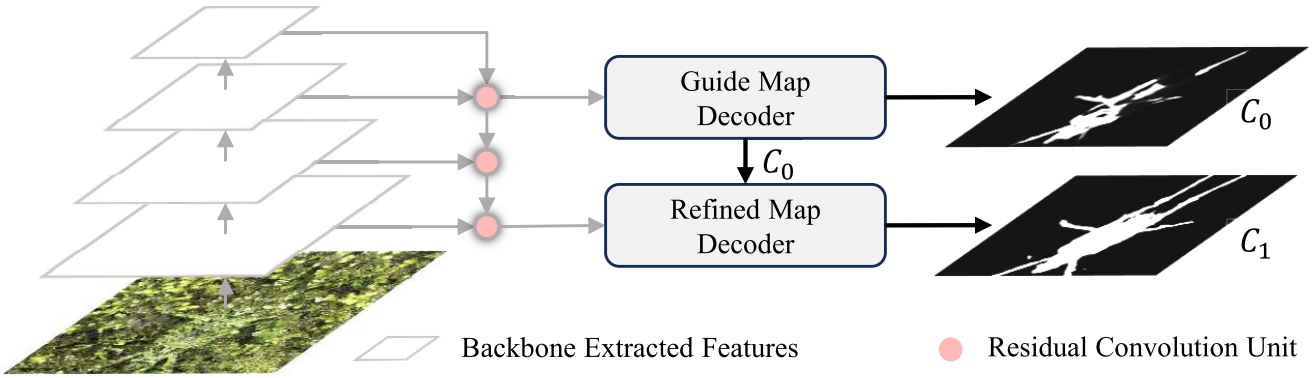


Fig. 2. Structure of the proposed difference-aware decoder, built upon multi-level features extracted from common backbones (*e.g.*, ResNet, Res2Net, PVT). It consists of two stages: the Guide Map Decoder (GMD) for coarse prediction and the Refined Map Decoder (RMD) for fine-grained enhancement.

feature pyramid networks (FPNs), progressively fusing multi-level features. In FPN, bilinear interpolation is typically used, but it can result in feature misalignment, which has led several studies [44], [45] to propose learning 2D transformation offsets among features at different levels. These transformations help to achieve more precise alignment, improving the fusion of multi-level features. In addition, Zheng et al. [46] introduced global representation embedding into FPNs to enhance low-level features, aiming to strengthen the discriminative power of features from different layers. Furthermore, Wang et al. [47] proposed modifications to the non-local self-attention mechanism to facilitate explicit interactions among layers, implementing a cross-level attention mechanism that aims to capture long-range dependencies between different levels.

Implicit feature fusion often lacks sufficient high-level semantic enhancement, limiting the separability of low-level features in binary segmentation. While explicit strategies like non-local attention [47], [48] can model global dependencies, they struggle to capture localized cues such as edges and textures. In contrast, our method employs a multi-scale local window mechanism to explicitly inject high-level semantics into low-level details, effectively enhancing fine-grained discriminability with low computational cost.

C. Window-Based Attention

Swin [11] was the first to introduce the shifted windowing strategy, which divides input tokens into non-overlapping windows and performs self-attention within each window. This approach achieves a trade-off between model performance and computational efficiency. Inspired by this window-based paradigm, numerous recent window-based attention mechanisms have emerged. For example, [49] introduced a large window attention that captures the contextual information at multiple scales, while [50] proposed Varied-Size Window Attention (VSA) to learn adaptive window configurations from the input data.

However, existing window-based strategies typically adopt independent single-layer self-attention in the encoder layers without consideration of the cross-level information exchange. Additionally, the shifted window partition ignores the relationships among windows, resulting in the incoherence of the detected results. Moreover, the effectiveness of multi-scale windows in cross-layer attention has not been thoroughly explored. Therefore, cross-level multi-scale window-based attention integrating the high-level semantics into the low-level details is essential for achieving high separability of foreground objects but is currently lacking in existing methods.

III. Methodology

A. Overview

This section describes the details of the proposed difference-aware decoder, as illustrated in Fig. 2. Our approach emulates the way human vision detects objects, leveraging multi-level features from the backbone. We employ a two-stage method to mimic the human visual system's process of detecting and refining the perception of objects [26]. In Stage A (Guide Map Decoder), high-level encoded features are used to create a guide map representing coarse foreground information. In Stage B (Refined Map Decoder), the guide map helps high-level features provide explicit semantic guidance to low-level features, enhancing the distinction between foreground and background and resulting in a more accurate segmentation map.

B. Stage A: Guide Map Decoder

In the first stage, we simulate the initial phase of human visual perception, where the brain conducts a preliminary scan of an image to achieve a rough detection of foreground objects. This rough detection is informed by the accumulated knowledge of object attributes, which in segmentation tasks is captured by the high-level features generated by the encoder. Given an input RGB image $I_m \in \mathbb{R}^{3\times H\times W}$, we extract encoded features from four layers, denoted as $E_i \in \mathbb{R}^{B\times C_i\times H_i\times W_i}$, where $H_i = \frac{H}{2^{i+1}}$ and $W_i = \frac{W}{2^{i+1}}$. These features are categorized into low-level features (E_1 and E_2) and high-level features (E_3 and E_4).

First, the features E_1, E_2, E_3, and E_4 are processed through a series of 3×3 convolutional layers to compress and standardize their channel dimensions:

$$E_i^c = \mathrm{Conv}_{3\times3}(E_i), \quad i = 1, 2, 3, 4 \tag{1}$$

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Here, $\text{Conv}_{3\times3}$ denotes a 3×3 convolution that unifies the feature channels to a unified dimension C, ensuring uniformity across different scales while also reducing the parameter count and computational complexity. These unified features are then passed through receptive field (RF) [13] blocks, designed to capture multi-scale contextual information through dilated convolutions with varying kernel sizes and dilation rates:

$$E_i^r = \text{RF}(E_i^c), \quad i = 1, 2, 3, 4 \tag{2}$$

Next, the encoded features E_3^r and E_4^r are fused to produce the high-level features F_h. This fusion step combines the features from the two layers, refining the high-level representation to better capture important semantic information:

$$F_h = \text{RCU}(\text{Up}^2(E_4^r) + \text{RCU}(E_3^r)), \tag{3}$$

where Up^2 represents the upsampling operation with a scale factor of 2, and RCU is the Residual Convolution Unit, defined as:

$$\text{RCU}(X) = X + \phi(\text{Conv}_{3\times3}(\phi(\text{Conv}_{3\times3}(X)))), \tag{4}$$

where ϕ denotes the activation function.

Finally, the output F_h is passed through a final 3×3 convolutional layer to generate the guide map C_0:

$$C_0 = \text{Conv}_{3\times3}(F_h). \tag{5}$$

This guide map serves as an initial coarse prediction of the target regions within the input image, guiding subsequent stages to focus on areas likely containing the object of interest.

C. Stage B: Refined Map Decoder

In Stage B, inspired by the human visual system's refinement process, we aim to enhance the separation between foreground and background by improving the discriminability of low-level details. This is accomplished via two key modules: (i) Difference-Aware Prototype Generation, which extracts representative prototypes for foreground and background to guide refinement, and (ii) Overlapped Window Cross-Level Semantic Guidance, which injects high-level semantic information into low-level features through an efficient attention mechanism.

The low-level features F_l are computed as:

$$F_l = \text{Up}^2\left(\text{RCU}(\text{Up}^2(F_h) + \text{RCU}(E_2^r))\right) + \text{RCU}(E_1^r). \tag{6}$$

1) Difference-Aware Prototype Generation: To leverage the guide map C_0 effectively, we adopt a prototype learning strategy inspired by robust representation learning in few-shot learning [51] and out-of-distribution detection [52]. While these prior works [48], [52], [53] primarily focus on learning noise-tolerant prototypes for classification or detection tasks, our approach adapts the idea to binary segmentation by introducing dynamic weighting to construct foreground and background prototypes. The resulting similarity maps generate foreground-aware and background-aware feature branches, which are further integrated through the Overlapped Window Cross-Level Semantic Guidance. This design enables precise refinement of low-level features F_l by aligning them with representative vectors derived from high-level features F_h and C_0, thereby mitigating guide map noise while remaining tailored to segmentation.

We first binarize C_0 into a mask M by applying sigmoid normalization followed by thresholding at 0.5:

$$M_{i,j} = \begin{cases} 1, & \sigma(C_{0,(i,j)}) \geq 0.5, \\ 0, & \sigma(C_{0,(i,j)}) < 0.5, \end{cases} \tag{7}$$

where $C_{0,(i,j)}$ denotes the predicted probability at pixel (i, j).

Next, we generate an intensity map I by summing channel-wise activations of F_h and normalizing to the range $[0, 1]$:

$$I_{i,j} = \frac{\sum_{c=1}^{C} F_h^{(c,i,j)}}{\max_{i,j}\left(\sum_{c=1}^{C} F_h^{(c,i,j)}\right)}. \tag{8}$$

Using I and M, we compute foreground and background prototypes P_f and P_b as weighted averages of F_h:

$$\begin{aligned} P_f &= \frac{\sum_{i,j} F_{h,(i,j)} \cdot I_{i,j} \cdot M_{i,j}}{\sum_{i,j} I_{i,j} \cdot M_{i,j} + \epsilon}, \\ P_b &= \frac{\sum_{i,j} F_{h,(i,j)} \cdot \left(1 - I_{i,j} \cdot M_{i,j}\right)}{\sum_{i,j} \left(1 - I_{i,j} \cdot M_{i,j}\right) + \epsilon}. \end{aligned} \tag{9}$$

To refine F_l, cosine similarity maps between F_l and these prototypes are calculated:

$$\begin{aligned} S_f &= \text{CosSim}(F_l, P_f), \\ S_b &= \text{CosSim}(F_l, P_b), \end{aligned} \tag{10}$$

where $\text{CosSim}(\cdot)$ denotes the cosine similarity operation measuring feature alignment.

These similarity maps modulate the low-level features, producing foreground-aware and background-aware features:

$$F_f = F_l \cdot S_f, \quad F_b = F_l \cdot S_b. \tag{11}$$

Subsequently, high-level semantic information encoded in F_h is explicitly leveraged to further refine both F_f and F_b. This refinement enhances the network's capability to distinguish foreground from background, especially in complex scenes with intricate or visually similar backgrounds. By combining detailed low-level features with rich semantic guidance, the model achieves more accurate segmentation, even under challenging or camouflaged conditions.

2) Overlapped Window Cross-Level Semantic Guidance: Given two input features, $F_h \in \mathbb{R}^{B\times C\times H_3\times W_3}$ (high-level semantic features) and $F_l \in \mathbb{R}^{B\times C\times H_1\times W_1}$ (low-level detail features, either F_f or F_b), our goal is to explicitly inject semantic guidance from F_h into F_l, enhancing low-level representations with rich high-level context. This is achieved through a cross-level attention module characterized by channel grouping, multi-scale overlapped window processing, and efficient linear attention.

To align features spatially, we compute the scale factor between resolutions:

$$S_s = \frac{H_1}{H_3}. \tag{12}$$

a) Channel grouping: To balance semantic specialization and computational efficiency, we partition both F_h and F_l into G groups along the channel dimension. Each group captures complementary subspace semantics with reduced complexity:

$$F_{h,g}, F_{l,g} = \text{Group}(F_h, F_l, g), \quad g \in \{1, \ldots, G\}. \tag{13}$$

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b) Multi-scale overlapped window unfolding: To enable explicit cross-level feature interaction, we propose a multi-scale overlapped window unfolding strategy that partitions high-level features $F_h \in \mathbb{R}^{B\times C\times H_h\times W_h}$ and low-level features $F_l \in \mathbb{R}^{B\times C\times H_l\times W_l}$ into spatially aligned local windows. The alignment is ensured by a scaling factor $S_s = \frac{H_l}{H_h}$, which guarantees one-to-one correspondence between high- and low-level regions.

Given a window size k_i, we unfold the grouped high-level feature $F_{h,g}$ and low-level feature $F_{l,g}$ as:

$$\text{Unfold}\left(F_{h,g}, k_i, \tfrac{k_i}{2}\right) = \{W_{h,g}^{i,j}\}_{j=1}^{m_i^2}, \tag{14}$$

$$\text{Unfold}\left(F_{l,g}, S_s\cdot k_i, S_s\cdot \tfrac{k_i}{2}\right) = \{W_{l,g}^{i,j}\}_{j=1}^{m_i^2}, \tag{15}$$

$$m_i = \left\lfloor \frac{H_h - k_i}{k_i/2} \right\rfloor + 1, \tag{16}$$

where overlapping is achieved via a stride of $\frac{k_i}{2}$, and each high-level window $W_{h,g}^{i,j} \in \mathbb{R}^{B\times C\times k_i\times k_i}$ is spatially aligned with its corresponding low-level window $W_{l,g}^{i,j} \in \mathbb{R}^{B\times C\times (S_s k_i)\times (S_s k_i)}$.

For example, when $F_h \in \mathbb{R}^{B\times C\times 30\times 30}$, $F_l \in \mathbb{R}^{B\times C\times 120\times 120}$, and $k_i = 8$, we obtain 36 aligned window pairs with 8×8 patches from F_h and 32×32 patches from F_l, enabling spatially consistent semantic guidance from high- to low-level features.

c) Multi-head linear attention: Each pair of windows $(W_{l,g}^{i,j}, W_{h,g}^{i,j})$ is flattened and projected for attention computation:

$$T_{l,g}^{i,j} = \text{LayerNorm}(\text{Flatten}(\text{Transpose}(W_{l,g}^{i,j}))), \tag{17}$$

$$T_{h,g}^{i,j} = \text{LayerNorm}(\text{Flatten}(\text{Transpose}(W_{h,g}^{i,j}))), \tag{18}$$

followed by projection into query, key, and value:

$$q_g^{i,j} = \text{Linear}_q(T_{l,g}^{i,j}), k_g^{i,j} = \text{Linear}_k(T_{h,g}^{i,j}), v_g^{i,j} = \text{Linear}_v(T_{h,g}^{i,j}), \tag{19}$$

with attention output:

$$z_g^{i,j} = \frac{1}{q_g^{i,j}(k_g^{i,j})^T + \epsilon}, \tag{20}$$

$$k_g^{i,j}v_g^{i,j} = \frac{1}{\sqrt{ws}}(k_g^{i,j})^T v_g^{i,j}, \tag{21}$$

$$O_g^{i,j} = \frac{q_g^{i,j}(k_g^{i,j}v_g^{i,j})}{z_g^{i,j}}. \tag{22}$$

d) Adaptive feature folding: We project the attention output back to the original spatial layout using a folding operation:

$$O_{\text{proj},g}^{i,j} = \text{Linear}_{\text{out}}(O_g^{i,j}), \tag{23}$$

$$F_g^i = \text{Fold}\left(O_{\text{proj},g}^{i,j}, S_s\cdot k_i, S_s\cdot \tfrac{k_i}{2}\right). \tag{24}$$

Outputs from different scales are fused via learnable weights:

$$F_g' = \sum_i \alpha_i \cdot F_g^i. \tag{25}$$

Then, group-wise features are aggregated via channel concatenation:

$$F_l' = \text{Concat}(F_g'). \tag{26}$$

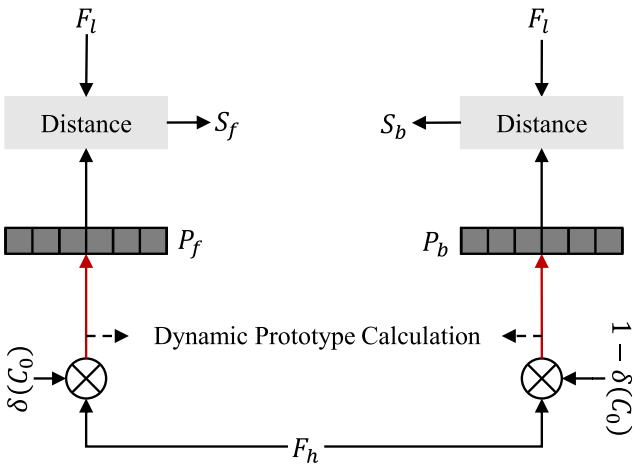


Fig. 3. The architecture of the Difference-Aware prototype generation method.

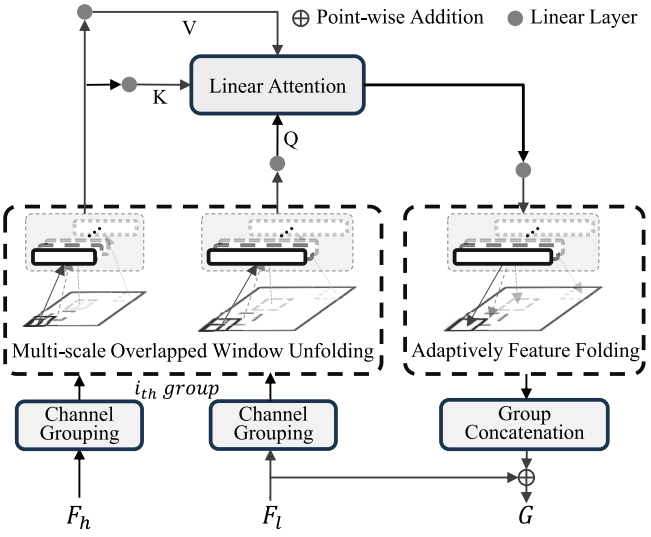


Fig. 4. The architecture of the overlapped window Cross-Level semantic guidance mechanism.

e) Weighted fusion: To preserve both the enhanced semantics and fine-grained details, we fuse F_l' with the original low-level features F_l using a learnable weight α:

$$G = \alpha\cdot F_l' + (1-\alpha)\cdot F_l. \tag{27}$$

This entire attention mechanism is applied independently to the prototype-aware foreground and background features F_f and F_b, resulting in enhanced features G_f and G_b. These are concatenated and decoded by a convolutional layer to produce the final prediction:

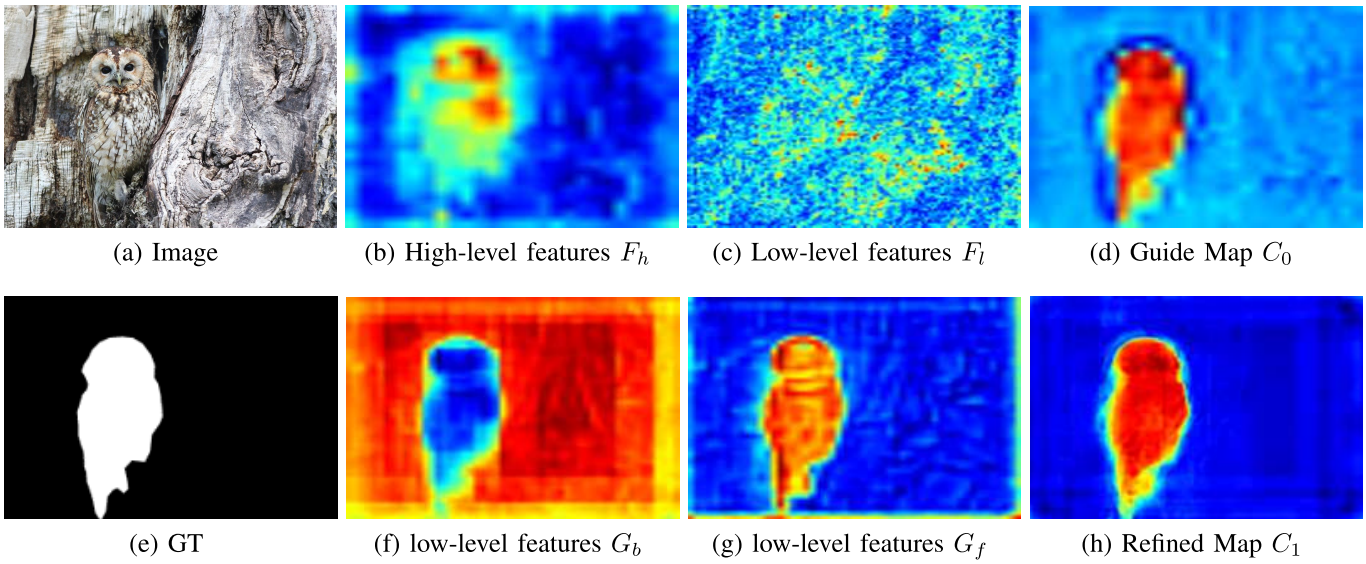
$$C_1 = \text{Conv}_{3\times 3}(\text{Concat}(G_f, G_b)). \tag{28}$$

D. Loss Function

In this paper, weighted binary cross-entropy (BCE) loss and weighted intersection over union (IoU) loss are combined as $loss = loss_{wbce} + loss_{wiou}$ for each output. The total loss is:

$$loss = loss(C_0, GT) + loss(C_1, GT), \tag{29}$$

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(a) Image (b) High-level features F_h (c) Low-level features F_l (d) Guide Map C_0

(e) GT (f) low-level features G_b (g) low-level features G_f (h) Refined Map C_1

Fig. 5. The feature visualization of each stage in the proposed method.

where $GT \in \{0,1\}^{B\times H\times W}$ denotes the binary ground truth mask with foreground labeled as 1 and background as 0. Both outputs C_0 and C_1 are upsampled to the original spatial resolution $H \times W$ before loss computation.

E. Analysis of the Two Stages

To validate the effectiveness of the proposed paradigm, we visualized the feature maps at each stage using real-world data. As shown in Fig.5 (b) and (c), the high-level features F_h struggle to capture fine details of the foreground objects, while the low-level features F_l fail to clearly distinguish the foreground from the background. Additionally, the generated guide map provides only a coarse segmentation result. However, in the second stage, the proposed Difference-aware Decoder effectively leverages the distinctions between the foreground and background. As illustrated in Fig.5 (f) and (g), the binary object generated using the fused difference-aware features is significantly clearer than the coarse map, demonstrating the clear advantage of the proposed paradigm (see Fig. 5 (h)).

IV. Experiments

In this section, we provide a detailed analysis of the experimental results across the Salient Object Detection (SOD) and Camouflaged Object Detection (COD).

A. Experimental Setting

In this study, we aimed to design a unified decoder network to capture and enlarge the difference between the foreground and the background, for binary segmentation. To evaluate the proposed model, experiments were conducted in SOD and COD. All the models were implemented in PyTorch 1.7.0, and the training and testing were achieved using an NVIDIA GeForce RTX 4090 GPU with 24 GB memory. The spatial size of the input images was resized to 480×480. The Adam optimizer was used in the training process. The initial learning rate was 10^{-4}, which was adjusted according to the formula $\left(1-\frac{epoch}{max_epoch}\right)^{power}$, where $power = 0.9$. It is also worth mentioning that the difference-aware decoder was trained for 40 epochs and the batch size was 8.

B. Evaluation Metrics

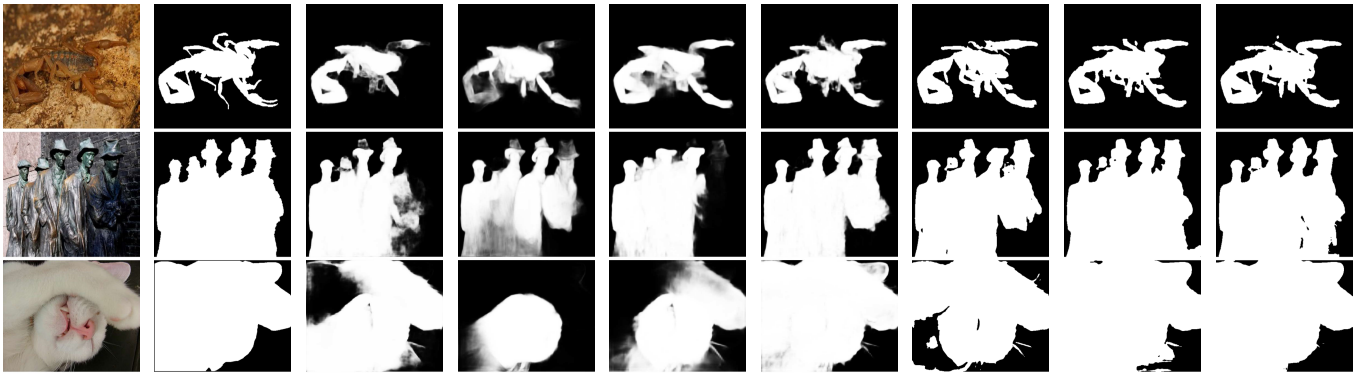
Following the previous works, we utilize the structure-measure (S_α), E-measure (E_ϕ), weighted F-measure (F_β^w) and mean absolute error (M) as the evaluation metrics.

C. Experiments in SOD

1) Datasets: We kept the same experimental setup as used for previous SOD methods [18]. In the experiments, the DUTS-TR [54] set was used for the training, and DUTS-TE [54] set, DUT-OMRON dataset [55], PASCAL-S dataset [56], HKU-IS dataset [57], ECSSD dataset [58], and SOD dataset [59] were used for the testing.

2) Compared Methods: For the SOD task, we selected 18 methods to evaluate the proposed difference-aware decoder: methods based on the ResNet-50 backbone, *i.e.*, PoolNet + (ResNet-50) [60], DFI (ResNet-50) [61], PSG (ResNet-50 [62]), MSFNet (ResNet-50) [63], ICON (ResNet-50) [64]; methods based on the Res2Net-50, ResNet-34, and ResNet-101 backbone, *i.e.*, CSF (Res2Net-50) [65], BASNet (ResNet-34) [18], GateNet (ResNet-101) [66], PoolNet (Res2Net-50) [60]; and a method based on the transformer (Swin Transformer, PVT-v2-b2) backbone, *i.e.*, PVT-SOD (PVT-v2-b2) [67], TIGAN (Swin Transformer) [68], SelfReformer (PVT-v2-b2) [69], and ICON(PVT-v2-b4) [64]. We directly used the results or the code provided by the related authors for the comparison. The proposed difference-aware decoder was tested on the VGG-16, ResNet-50, Res2Net-50, and PVT-v2-b2 backbones, for a fair comparison. It is worth noting that PVT-v2-b2 only has a four-layer encoder structure. Therein, for the proposed DAD, we select the first and fourth layer features for Stage A; whereas the first, second and third layers for Stage B.

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(a) Image (b) GT (c) MINet-R (d) Pool+-R (e) ICON-R (f) ICON-P (g) **DAD-R** (h) **DAD-R2** (i) **DAD-P**

Fig. 6. Visual SOD results. R, R2, and P represent the ResNet-50, Res2Net-50, and PVT-v2-b2, respectively.

TABLE I

PERFORMANCE COMPARISON WITH BASELINE MODELS ON DUTS-TE, HKU-IS, PASCAL-S, DUT-OMRON, SOD, AND ECSSD DATASETS. ↑ INDICATES THE HIGHER THE SCORE THE BETTER AND VICE VERSA. THE BEST SCORE OF EACH BACKBONE IS HIGHLIGHTED IN BOLD, WHILE THE SECOND-BEST SCORE FOR EACH METRIC IS UNDERLINED

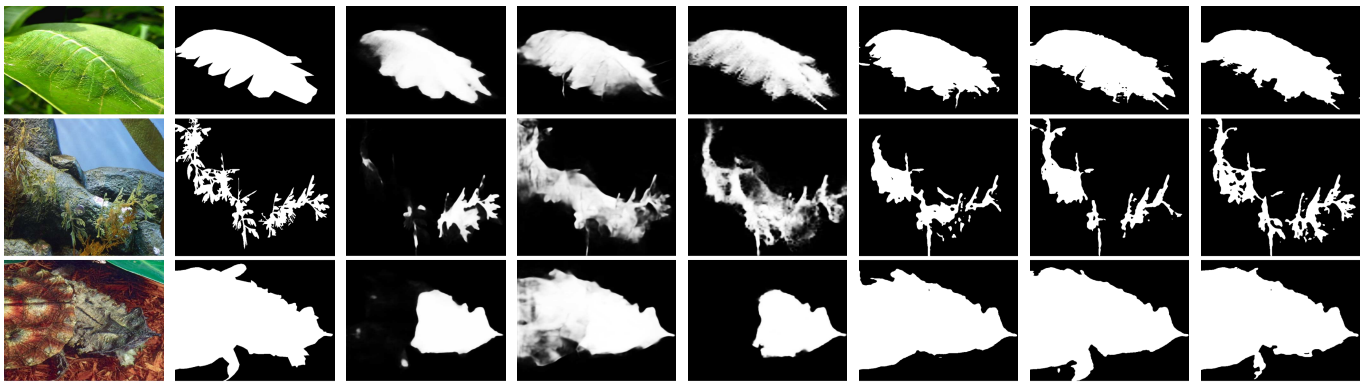
Baseline	Backbone	DUTS-TE		HKU-IS		PASCAL-S		DUT-OMRON		SOD		ECSSD	
		$E_\phi\uparrow$	$F_\beta^w\uparrow$	$E_\phi\uparrow$	$F_\beta^w\uparrow$	$E_\phi\uparrow$	$F_\beta^w\uparrow$	$E_\phi\uparrow$	$F_\beta^w\uparrow$	$E_\phi\uparrow$	$F_\beta^w\uparrow$	$E_\phi\uparrow$	$F_\beta^w\uparrow$
PoolNet+	ResNet-50	0.910	0.817	-	-	0.897	0.801	0.841	0.716	0.805	**0.804**	0.945	0.902
DFI	ResNet-50	0.912	0.817	0.948	0.890	0.901	0.814	0.862	0.738	0.822	0.764	0.947	0.906
PSG	ResNet-50	0.919	0.831	0.947	0.891	0.892	0.807	0.865	0.747	-	-	0.947	0.910
MSFNet	ResNet-50	0.928	0.841	0.956	0.903	0.901	0.822	0.873	0.757	-	-	0.951	0.916
ICON	ResNet-50	0.924	0.836	0.953	0.902	0.899	0.818	0.876	0.761	**0.854**	0.794	**0.954**	0.918
DAD	ResNet-50	**0.941**	**0.871**	**0.960**	**0.915**	**0.910**	**0.834**	**0.886**	**0.781**	0.837	0.786	**0.954**	**0.925**
CSF	Res2Net-50	0.912	0.823	0.947	0.891	0.889	0.807	0.854	0.734	0.820	0.757	0.949	0.910
BASNet	ResNet-34	0.907	0.825	0.948	0.900	0.878	0.808	0.868	0.760	0.822	0.762	0.947	0.912
GateNet	ResNet-101	0.912	0.821	-	-	0.888	0.803	0.858	0.727	-	-	0.943	0.907
PoolNet	Res2Net-50	0.902	0.804	0.942	0.881	0.884	0.800	0.857	0.732	0.815	0.752	0.946	0.906
DAD	Res2Net-50	**0.944**	**0.871**	**0.963**	**0.917**	**0.920**	**0.844**	**0.900**	**0.799**	**0.860**	**0.808**	**0.959**	**0.927**
PVT-SOD	PVT-v2-b2	0.933	0.865	**0.973**	**0.941**	0.906	0.851	0.883	0.791	-	-	0.958	0.933
TIGAN	Swin	0.941	-	0.962	-	0.916	-	0.890	-	-	-	0.964	-
SelfReformer	PVT-v2-b2	0.920	0.872	0.960	0.915	0.873	0.848	0.886	0.783	-	-	0.929	0.926
ICON	PVT-v2-b4	0.950	0.882	0.967	0.925	0.921	0.847	0.896	0.793	**0.864**	0.813	0.964	0.933
DAD	PVT-v2-b2	**0.957**	**0.904**	0.971	0.936	**0.922**	**0.858**	**0.914**	**0.831**	0.861	**0.816**	**0.967**	**0.945**

3) Visual Results of the SOD: To compare the visual effects of different SOD methods, we present several salient images in Fig. 6. The difference-aware decoder consistently outperforms other state-of-the-art SOD methods. In the first image, accurately segmenting the scorpion's pincers is challenging. Using ResNet-50 as the backbone, GateNet, MINet, and PoolNet++ exhibit significant omissions, while ICON captures the general outline but is coarse. In contrast, our DAD achieves precise segmentation of the scorpion's body across all backbones—ResNet-50, Res2Net-50, and PVT-v2-b2. In the second image, five individuals are visible. With ResNet-50, both GateNet and ICON-R fail to achieve complete detection, while MINet and PoolNet+ perform less effectively than our DAD. When using PVT-v2-b2, ICON cannot segment the lower half of the person on the far right, whereas our method successfully does so. In the third image, for the salient cat body, GateNet, MINet, PoolNet++, and ICON-R miss substantial portions. Although ICON-P captures the outline, it lacks the refinement of our DAD. These examples demonstrate the effectiveness of the difference-aware decoder in achieving fine-grained segmentation by enhancing foreground-background separability in low-level details.

4) Quantitative Evaluation of the SOD: Table I lists the overall quantitative evaluation results of the different methods obtained on the six datasets. The best score for each backbone is highlighted in bold, and the second score for each metric is underlined. The proposed difference-aware decoder, when using the PVT-v2-b2 backbone, achieves the top performance across all datasets and evaluation metrics, except for the HKU-IS dataset, where it ranks second. Futhermore, with the different backbones of ResNet-50, and Res2Net-50, the difference-aware decoder always achieves the best performance, indicating the transferability of the proposed decoder architecture for different backbones.

Compared to the other methods, the difference-aware decoder improves the evaluation metrics by a large margin. The main reason for this advantage is that the proposed decoder effectively amplifies the distinction between the

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(a) Image (b) GT (c) Zoom-R (d) FAP-R2 (e) UEDG-P (f) **DAD-R** (g) **DAD-R2** (h) **DAD-P**

Fig. 7. Visual COD results. R, R2, and P represent the ResNet-50, Res2Net-50, and PVT-v2-b2, respectively.

foreground and background by leveraging a two-stage strategy, which enhances the discriminability of low-level details.

D. Experiments in COD

1) Datasets: Following the previous COD works, four datasets were used to evaluate the proposed model: CHAMELEMON [70], CAMO [71], COD10K [2], NC4K [72]. According to [2], [22], [73], we used the training images from the CAMO and COD10K datasets as the training set (4,040 images) and the test images from the CHAMELEMON [70], CAMO [71], COD10K [2], and NC4K [72] as the test set.

2) Compared Methods: In order to evaluate the proposed difference-aware decoder, we selected 16 state-of-the-art methods for comparison: methods based on the ResNet-50 backbone, *i.e.*, PFNet (ResNet-50) [22], SegMaR (ResNet-50) [33], ZoomNet (ResNet-50) [28], DINet [74]; methods based on the Res2Net-50 backbone, *i.e.*, SINet V2 (Res2Net-50) [73], BGSNet [16], FAPNet [30], SFCNet [75], and DSNet [76]; and methods based on the transformer backbone, *i.e.*, UEDG [19], VSCode [77], FSPNet [78], RISNet [79], SDRNet [80], CamoFormer [81], PNet [82], FocusDiffuser [24], and CamoDiffusion [25]. The proposed difference-aware decoder was tested on the ResNet-50, Res2Net-50, and transformer (PVT-v2-b2) backbones, for a fair comparison.

3) Visual Results of the COD: To compare the visual performance of various methods, we selected several challenging camouflaged images for visualization, as shown in Fig. 7. In the first image, the insect camouflages by blending into the leaves with a similar color, making it difficult for SegMaR, ZoomNet, and DAD with the ResNet-50 backbone to fully detect all parts of the insect. In contrast, our method achieves contour extraction closest to the ground truth, particularly when using Res2Net-50 and PVT-v2-b2 as the backbone. In the second image, the key challenge is detecting the camouflaged branches and leaves of the plant. Notably, with ResNet-50 as the backbone, our DAD provides more complete segmentation than SegMaR and ZoomNet. Additionally, with Res2Net-50, our DAD surpasses SINet v2 and FAPNet. This advantage becomes even more pronounced with PVT-v2-b2, allowing for clear and precise detection of the plant's contours. In the third image, the most challenging aspect is fully detecting the left side of the animal's body. Aside from FAPNet, none of the other methods correctly identifies this region. In contrast, our DAD demonstrates a significant advantage in accurately detecting the camouflaged body. These results highlight the critical role of the backbone in COD tasks. However, our proposed decoder paradigm optimally utilizes different backbones to achieve superior accuracy.

4) Quantitative Evaluation of the COD: Table II presents the overall quantitative evaluation results of the different methods across four datasets. Using the ResNet-50 backbone, the proposed difference-aware decoder achieves the highest score in the F_β^w metric, as well as the best scores in E_ϕ and M on the CAMO dataset. It also ranks second in E_ϕ, F_β^w, and M on the COD10K dataset, and secures the top scores in E_ϕ, F_β^w, and M, while ranking second in S_α on the NC4K dataset. When using Res2Net-50, PVT-v2-b2 and PVT-v2-b2 backbones, our method consistently outperforms the compared approaches across almost all metrics and datasets by a significant margin. This advantage can be attributed to the proposed difference-aware decoder, which enhances the distinction between camouflaged objects and their background. By employing a two-stage strategy, it effectively amplifies subtle differences, improving the DAD's ability to detect these challenging objects.

V. Discussion

In order to verify the effectiveness of the parameter selection in each module in the experiments, we used ResNet-50 as the backbone and selected two COD datasets (COD10K [2], NC4K [72]) and two SOD datasets (DUTS-TE [54], HKU-IS [57]) for the validation experiments.

A. Ablation Study

1) Guide Map Versus Refined Map: In this study, we evaluate the performance of our proposed dual-branch decoder paradigm by comparing the precision of the Guide Map and the Refined Map across several benchmark datasets. The results, presented in Table III, underscore the effectiveness of our approach in enhancing object segmentation accuracy.

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TABLE II
PERFORMANCE COMPARISON WITH BASELINE MODELS ON COD DATASETS

Baseline	Backbone	CAMO				CHAMELLON				COD10K				NC4K			
		$S_\alpha\uparrow$	$E_\phi\uparrow$	$F_\beta^w\uparrow$	$M\downarrow$	$S_\alpha\uparrow$	$E_\phi\uparrow$	$F_\beta^w\uparrow$	$M\downarrow$	$S_\alpha\uparrow$	$E_\phi\uparrow$	$F_\beta^w\uparrow$	$M\downarrow$	$S_\alpha\uparrow$	$E_\phi\uparrow$	$F_\beta^w\uparrow$	$M\downarrow$
PFNet	ResNet-50	0.782	0.842	0.695	0.085	0.882	0.931	0.808	0.032	0.800	0.877	0.660	0.040	0.829	0.886	0.747	0.053
SegMaR	ResNet-50	0.815	0.872	0.742	0.071	**0.906**	0.954	**0.860**	0.025	0.833	0.895	0.724	0.033	0.841	0.907	0.781	0.046
ZoomNet	ResNet-50	**0.820**	**0.892**	0.752	**0.066**	0.902	**0.958**	0.845	**0.023**	**0.838**	**0.911**	**0.729**	**0.029**	**0.853**	0.912	0.784	0.043
DINet	ResNet-50	0.790	0.851	0.747	0.082	-	-	-		0.804	0.879	0.696	0.039	0.837	0.891	0.787	0.051
DAD	ResNet-50	0.809	0.890	**0.764**	0.069	0.876	0.943	0.831	0.028	0.823	0.908	0.727	0.030	0.847	**0.917**	**0.796**	**0.041**
SINet V2	Res2Net-50	**0.820**	0.882	0.743	0.070	0.888	0.942	0.816	0.030	0.815	0.887	0.680	0.037	0.847	0.903	0.767	0.048
BGSNet	Res2Net-50	0.796	0.851	0.717	0.079	**0.895**	0.946	0.841	0.027	0.818	0.891	0.699	0.034	0.814	0.897	0.771	0.048
FAPNet	Res2Net-50	0.815	0.865	0.734	0.076	0.893	0.940	0.825	0.028	0.822	0.888	0.694	0.036	0.851	0.899	0.775	0.047
SFCNet	Res2Net-50	0.808	0.850	0.732	0.073	0.876	0.921	0.807	0.032	0.814	0.873	0.695	0.035	0.839	0.883	0.767	0.049
DSNet	Res2Net-50	0.817	0.870	0.726	0.073	0.877	0.930	0.787	0.035	0.809	0.878	0.657	0.038	0.843	0.894	0.753	0.050
DAD	Res2Net-50	**0.820**	**0.895**	**0.781**	**0.065**	0.886	**0.960**	**0.847**	**0.025**	**0.836**	**0.918**	**0.750**	**0.028**	**0.854**	**0.920**	**0.807**	**0.039**
UEDG	PVT-v2-b2	0.868	0.922	0.819	0.048	**0.911**	**0.958**	**0.866**	0.023	0.858	0.924	0.766	0.025	0.881	0.928	0.829	0.035
VSCode	Swin	0.836	0.892	0.768	0.060	-	-	-	-	0.847	0.913	0.744	0.028	0.874	0.920	0.813	0.038
FSPNet	ViT	0.856	0.899	0.799	0.050	0.908	0.943	0.851	0.023	0.851	0.895	0.735	0.026	0.879	0.915	0.816	0.035
RISNet	PVT-v2-b2	**0.870**	0.922	0.827	0.050	-	-	-	-	**0.873**	0.931	**0.799**	0.025	0.882	0.925	0.834	0.037
DAD	PVT-v2-b2	0.869	**0.933**	**0.845**	**0.043**	0.899	**0.958**	0.863	**0.021**	0.863	**0.936**	0.794	**0.021**	**0.884**	**0.939**	**0.852**	**0.029**
SDRNet	PVT-v2-b4	0.872	0.924	0.826	0.049	**0.914**	0.950	0.868	0.023	0.871	0.924	0.785	0.023	0.889	0.934	0.842	0.032
CamoFormer	PVT-v2-b4	0.872	0.931	0.831	0.046	-	-	-	-	0.869	0.931	0.786	0.023	0.892	0.941	0.847	0.030
FocusDiffuser	PVT-v2-b4	0.881	0.939	0.851	0.042	-	-	-	-	0.875	0.939	0.809	0.020	0.891	0.940	0.854	0.029
CamoDiffusion	PVT-v2-b4	0.878	0.940	0.853	0.042	-	-	-	-	**0.881**	0.944	0.814	0.020	**0.893**	0.942	0.859	0.029
DAD	PVT-v2-b4	**0.885**	**0.948**	**0.868**	**0.037**	0.912	**0.972**	**0.887**	**0.018**	0.880	**0.949**	**0.824**	**0.018**	**0.893**	**0.947**	**0.868**	**0.027**

means that the results were obtained by ourselves.

TABLE III
PERFORMANCE COMPARISON BETWEEN GUIDE MAP C_0 AND REFINED MAP C_1 ACROSS MULTIPLE DATASETS

Baseline	B	COD10K		NC4K		DUTS-TE		HKU-IS	
		$S_\alpha\uparrow$	$F_\beta^w\uparrow$	$S_\alpha\uparrow$	$F_\beta^w\uparrow$	$S_\alpha\uparrow$	$F_\beta^w\uparrow$	$S_\alpha\uparrow$	$F_\beta^w\uparrow$
Guide Map	R	0.799	0.688	0.808	0.738	0.879	0.848	0.903	0.900
Refined Map	R	**0.823**	**0.727**	**0.847**	**0.796**	**0.896**	**0.871**	**0.916**	**0.915**
Guide Map	R2	0.809	0.706	0.839	0.786	0.875	0.843	0.901	0.898
Refined Map	R2	**0.836**	**0.750**	**0.854**	**0.807**	**0.895**	**0.871**	**0.917**	**0.917**
Guide Map	P	0.840	0.757	0.869	0.832	0.903	0.886	0.918	0.919
Refined Map	P	**0.863**	**0.794**	**0.884**	**0.852**	**0.918**	**0.904**	**0.932**	**0.936**

These improvements are consistent across various backbone networks, including ResNet-50 (R), Res2Net-50 (R2), and PVT-v2-b2 (P). Specifically, when using ResNet-50, the Refined Map achieves an average increase of 0.023 in S_α and 0.034 in F_β^w over the Guide Map. This trend continues with Res2Net-50 and PVT-v2-b2, where the Refined Map consistently outperforms the Guide Map in terms of both metrics.

The superior performance of the Refined Map can be attributed to its capability to leverage multi-level features more effectively. By refining the coarse detections with detailed low-level feature exploration, our approach enhances the model's ability to discern subtle differences between foreground and background. This is evident in the significant performance gains observed, demonstrating the effectiveness of integrating high-level semantic information into low-level details.

2) Effectiveness of Prototype: The evaluation results presented in Table IV provide compelling evidence for the effectiveness of incorporating prototypes in our proposed DAD. In detail, the integration of prototypes into the DAD framework, derived from the Guide Map, markedly enhances the model's ability to differentiate between foreground and background. By computing cosine similarity between low-level features and the foreground/background prototypes, the model achieves more precise spatial alignment. This refinement results in significant performance improvements across datasets, as evidenced by higher S_α and F_β^w scores and lower M values. These results demonstrate that prototype-based guidance effectively enhances the model's discrimination capability, leading to more accurate and robust foreground extraction, especially in complex backgrounds.

3) Benefit from Cross-layer Interaction: As demonstrated in Table IV, the introduction of cross-layer interaction yields significant improvements in performance compared to the Baseline. Whether applied to foreground or background semantic guidance, the model exhibits substantial gains across various tasks and datasets. This enhancement can be attributed to our novel cross-level attention mechanism based on multi-scale local windows. By explicitly guiding low-level features to differentiate between foreground and background information, this approach effectively resolves previously challenging distinctions, resulting in marked performance improvements. Additionally, our findings reveal that incorporating both foreground and background prototypes simultaneously provides superior results compared to using a single prototype. This dual-prototype approach enables the model to more effectively capture subtle differences, leading to more accurate segmentation.

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TABLE IV
THE EVALUATION RESULTS ON SELECTED DATASETS. THE BEST SCORE FOR EACH METRIC IS HIGHLIGHTED IN BOLD

Settings					COD10K				NC4K				DUTS-TE				HKU-IS			
Baseline	Prototype		Guidance		$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^w \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^w \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^w \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^w \uparrow$	$M \downarrow$
	Fore	Back	Fore	Back																
✓					0.787	0.877	0.665	0.038	0.819	0.894	0.754	0.050	0.874	0.910	0.823	0.040	0.906	0.942	0.882	0.033
✓	✓		✓		0.805	0.892	0.690	0.035	0.833	0.905	0.771	0.046	0.879	0.923	0.839	0.035	0.908	0.949	0.899	0.030
✓		✓		✓	0.800	0.888	0.682	0.036	0.829	0.903	0.770	0.047	0.879	0.925	0.838	0.035	0.910	0.952	0.901	0.029
✓			✓	✓	0.815	0.900	0.707	0.032	0.838	0.911	0.785	0.044	0.888	0.927	0.850	0.033	0.905	0.953	0.900	0.028
✓	✓	✓	✓	✓	**0.823**	**0.908**	**0.727**	**0.030**	**0.847**	**0.917**	**0.796**	**0.041**	**0.896**	**0.941**	**0.871**	**0.029**	**0.916**	**0.960**	**0.915**	**0.025**

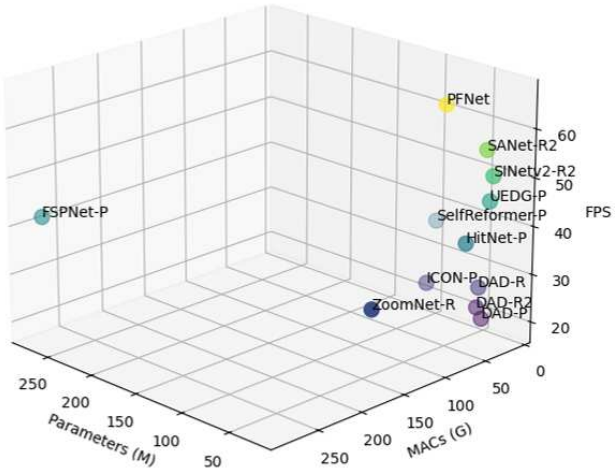


Fig. 8. Complexity and accuracy comparison of our DAD and the comparison methods.

TABLE V
COMPARISON OF DIFFERENT COMBINATIONS OF FEATURE DIVISION STRATEGIES

High	Low	COD10K		NC4K		DUTS-TE		HKU-IS	
		$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$S_\alpha \uparrow$	$F_\beta^w \uparrow$
4	1,2,3	0.801	0.688	0.830	0.769	0.880	0.838	0.905	0.900
3,4	**1,2**	**0.823**	**0.727**	**0.847**	**0.796**	**0.888**	**0.850**	**0.916**	**0.915**
2,3,4	1	0.799	0.688	0.808	0.738	0.885	0.854	0.909	0.906

4) Complexity Analysis: To more comprehensively evaluate the efficiency of our proposed method, we calculated the Multiply-Accumulate Operations (MACs), parameter count, and Frames Per Second (FPS), comparing these metrics against other state-of-the-art methods. As shown in Fig. 8, our approach exhibits a notable advantage in both parameter count and MACs, reflecting its computational efficiency. However, the two-stage process, designed to mimic human visual perception, involves multi-scale cross-window attention to explicitly guide low-level details using high-level semantics. This strategy enhances the foreground-background separability but comes at the cost of slower inference speed (FPS), making our model less competitive in this regard.

B. Parametric Analysis

1) Semantic Versus Detail Feature Division: As previously mentioned, our proposed DAD (Difference-aware Decoder) achieves precise binary segmentation by simulating the two stages of human visual perception. The first stage leverages attribute information, while the second stage focuses on enhancing the separability at the detail level. The encoder extracts four levels of features, with decreasing spatial dimensions and gradually enriched semantic information. To determine the optimal combination of features, we start by considering that all four encoded features must be utilized. We then arrange them into three different combinations, as shown in Table V. Among these combinations, it is evident that using the last two layers as high-level features for the first stage, and combining the upper two layers as low-level features for the second stage, yields the best results.

TABLE VI
PERFORMANCE METRICS FOR DIFFERENT NUMBERS OF CHANNEL GROUPS IN THE OVERLAPPED WINDOW CROSS-LEVEL SEMANTIC GUIDANCE MECHANISM ACROSS MULTIPLE DATASETS

G	COD10K		NC4K		DUTS-TE		HKU-IS		FLOPS(M)
	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	
1	0.805	0.698	0.839	0.784	0.881	0.847	0.905	0.900	174.152704
2	0.807	0.700	0.840	0.785	0.880	0.847	0.902	0.897	58.506752
4	**0.823**	**0.727**	**0.847**	**0.796**	**0.888**	**0.850**	**0.916**	**0.915**	25.566976
8	0.806	0.697	0.837	0.780	0.883	0.850	0.906	0.903	15.317888

2) Channel Grouping Impact: In this study, we adopt a channel grouping strategy with $G = 4$, which we found to be a favorable configuration through extensive experiments. This design brings two notable benefits: First, grouping channels into multiple semantic subspaces enables fine-grained decomposition of feature representations. This is particularly beneficial for complex foreground-background disentanglement, where localized semantic cues must be preserved and selectively guided. Compared to single-group baselines ($G = 1$), our grouped attention mechanism can better capture diverse semantics across spatial regions, leading to more discriminative foreground and background representations. Second, this strategy significantly reduces computational cost. By restricting attention computation within smaller channel groups, the total number of operations is greatly decreased. For instance, $G = 4$ results in approximately 10 times fewer FLOPs compared to the non-grouped setting, without sacrificing segmentation quality.

3) Window Size Sensitivity: Table VII presents the performance metrics for the Overlapped Window Cross-Level Semantic Guidance mechanism using various window sizes

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TABLE VII

PERFORMANCE METRICS FOR DIFFERENT WINDOW SIZES IN THE OVERLAPPED WINDOW CROSS-LEVEL SEMANTIC GUIDANCE MECHANISM ACROSS MULTIPLE DATASETS

W	COD10K		NC4K		DUTS-TE		HKU-IS		FLOPS(M)
	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	
2	0.804	0.695	0.799	0.743	0.886	0.854	0.908	0.905	**19.872000**
4	0.807	0.703	0.804	0.744	0.884	0.851	0.909	0.906	**19.872000**
6	0.809	0.706	0.803	0.708	0.881	0.851	0.901	0.902	**19.872000**
8	0.807	0.701	0.805	0.713	0.876	0.841	0.899	0.893	19.864576
(2,4)	0.806	0.698	0.818	0.738	0.883	0.850	0.906	0.901	21.772800
(2,6)	0.813	0.709	0.827	0.751	0.888	0.856	0.911	0.908	21.772800
(2,8)	0.804	0.695	0.817	0.734	0.880	0.846	0.904	0.899	21.765376
(4,6)	0.868	0.819	0.827	0.751	0.888	0.856	0.911	0.908	21.772800
(4,8)	0.789	0.673	0.820	0.740	0.885	0.853	0.909	0.905	21.765376
(2,4,6)	0.817	0.718	0.823	0.747	0.885	0.855	0.907	0.904	23.673600
(2,4,8)	0.809	**0.727**	0.829	0.755	**0.891**	**0.861**	0.912	0.909	23.666176
(2,4,6,8)	**0.823**	**0.727**	**0.847**	**0.796**	0.888	0.850	**0.916**	**0.915**	25.566976

TABLE VIII

PERFORMANCE METRICS FOR DIFFERENT OVERLAPPING RATIOS IN THE OVERLAPPED WINDOW CROSS-LEVEL SEMANTIC GUIDANCE MECHANISM ACROSS MULTIPLE DATASETS

Ratio	COD10K		NC4K		DUTS-TE		HKU-IS		FLOPS(M)
	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	
0	0.800	0.689	0.834	0.777	0.877	0.839	0.901	0.894	**19.864960**
1/2	**0.823**	**0.727**	**0.847**	**0.796**	**0.888**	**0.850**	**0.916**	**0.915**	25.566976

across multiple datasets. The results reveal how different window sizes impact both model performance and computational complexity.

For single window sizes, window sizes of 2 and 4 generally deliver competitive performance, with S_α and F_β^w scores being quite similar. Specifically, a window size of 4 slightly outperforms size 2 on COD10K and NC4K, while window size 6 shows a marginal improvement on DUTS-TE. Overall, the performance differences among these individual sizes are relatively modest.

In contrast, employing multi-window combinations tends to improve performance, suggesting that using a range of window sizes captures features at different scales more effectively, thus enhancing model accuracy. Among the multi-window combinations, the (2,4,6,8) setting achieves the highest overall performance with the highest S_α score across several datasets. However, this combination also results in increased computational complexity, with a FLOPS of 25.566976M, compared to the 19.872000M to 21.772800M observed for single or smaller combinations of window sizes.

In summary, multi-window combinations significantly enhance performance by explicitly guiding high-level semantic information to low-level details across diverse scales, improving foreground-background separability in complex COD and SOD scenarios. However, they incur higher computational costs. To balance performance and efficiency, we adopt the (2,4,6,8) window combination, leveraging its superior ability to integrate semantic guidance with fine-grained details despite increased computational demands.

4) Overlapping Ratio Effects: To assess the impact of overlapping ratios in the Overlapped Window Cross-Level Semantic Guidance mechanism, we conducted a detailed evaluation from both quantitative and qualitative perspectives.

A. Quantitative analysis: Table VIII illustrates that introducing overlapping windows results in considerable performance improvements across all datasets. Specifically, with an overlapping ratio of 1/2, the model shows enhanced accuracy metrics compared to the model without the overlapping strategy. For instance, S_α and F_β^w metrics increase notably for COD10K, NC4K, DUTS-TE, and HKU-IS datasets, indicating better segmentation performance with overlapping windows. This trend is consistent across all tested datasets, confirming that overlapping windows contribute to higher accuracy and improved feature differentiation.

B. Qualitative analysis: Fig. 10 provides visual insights into the effect of overlapping windows. The absence of overlap results in noticeable discontinuities in the segmentation outputs, reflecting the fragmented processing of the lowest-level features. In contrast, overlapping windows mitigate these issues, leading to more coherent and detailed segmentation results. This qualitative observation underscores the importance of using overlapping windows to preserve and enhance the internal details of objects, reinforcing the benefits highlighted in the quantitative results.

Overall, both quantitative and qualitative analyses validate the effectiveness of incorporating overlapping windows in the cross-level semantic guidance mechanism, demonstrating its critical role in improving segmentation performance and detail preservation.

5) Linear Attention Versus Softmax Attention: In selecting an attention mechanism for DAD, we evaluated softmax attention and linear attention for their ability to capture long-range dependencies critical for semantic-guided foreground-background disentanglement in COD and SOD tasks. Softmax attention normalizes weights to sum to 1, excelling at capturing fine-grained details and complex relationships. However, its quadratic complexity ($O(n^2)$) incurs significant computational overhead for high-resolution features or multi-scale operations. In contrast, linear attention reduces complexity to $O(n)$, enabling efficient processing of large inputs while maintaining robust dependency modeling.

In our module, we opted for linear attention to balance performance with efficiency, particularly in conjunction with a multi-scale window strategy. This approach allows us to manage larger inputs and more complex configurations without a substantial increase in computational cost. Although softmax attention may be better for detailed feature extraction, linear attention has proven to be a highly effective and efficient alternative in our experiments.

C. Robustness to Guide Map Perturbations

To evaluate the sensitivity of the Refined Map Decoder to the quality of the Guide Map, we conducted experiments on the COD benchmark by introducing random occlusions to the foreground at ratios of 10%, 20%, 30%, and 40%. Performance was measured using S_α (Fig. 11).

The results indicate that for occlusion ratios below 20%, performance degradation is minor, demonstrating robustness

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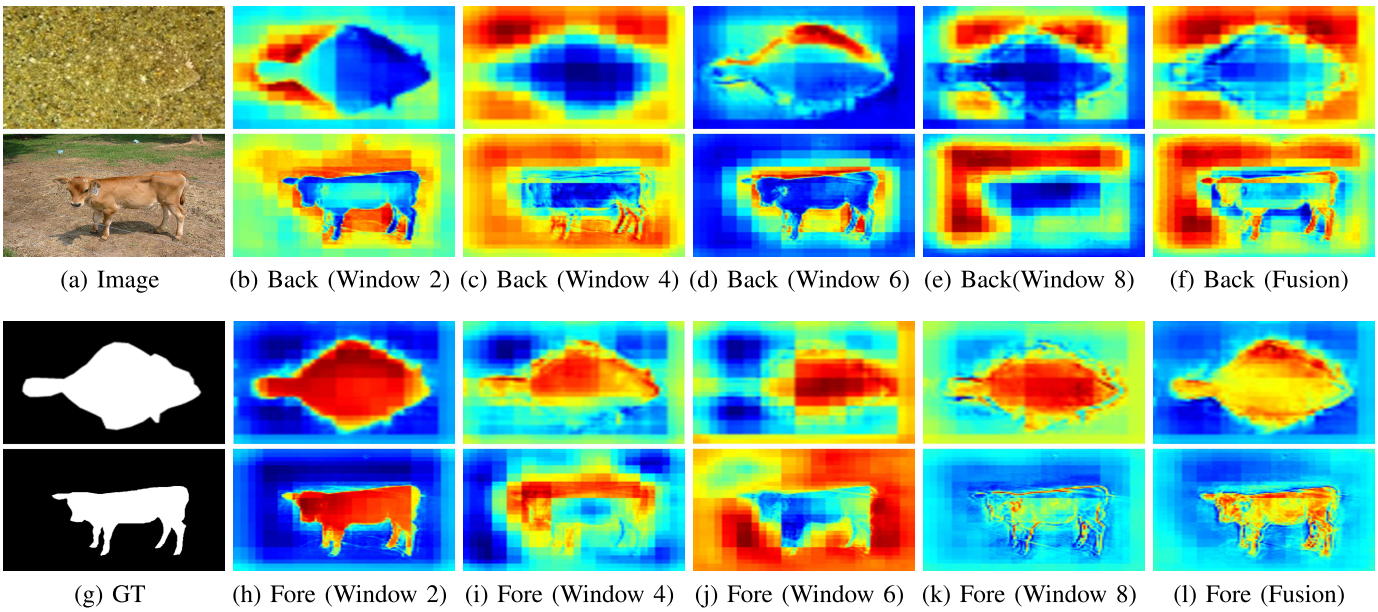


Fig. 9. Visualization of feature maps generated using different window sizes in the overlapped window Cross-Level semantic guidance mechanism.

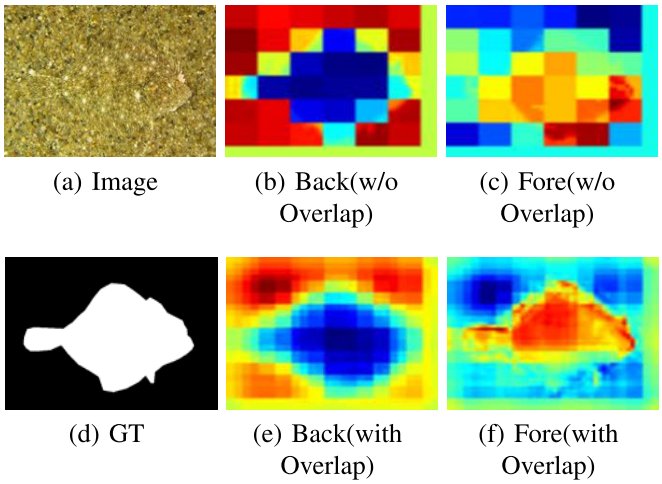


Fig. 10. Visual comparison of features (foreground and background) with and without overlapping.

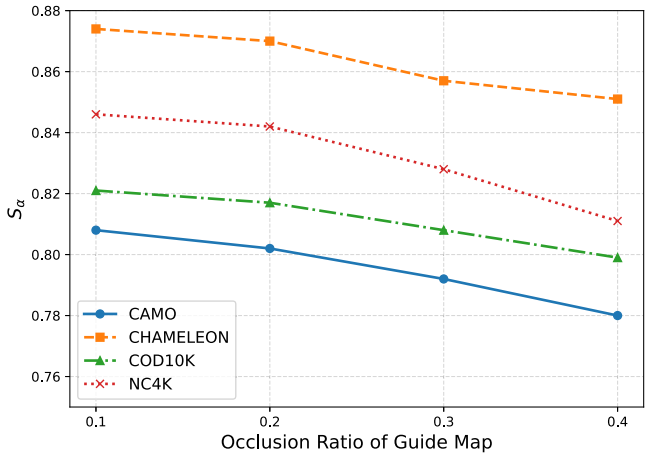


Fig. 11. Effects of random occlusions on the guide map for different COD datasets.

to moderate guide map noise. This resilience can be attributed to two key components: (1) the **Difference-Aware Prototype Generation** module, which constructs global prototypes to mitigate sensitivity to local errors; and (2) the **Overlapped Window Cross-Level Semantic Guidance** mechanism, which injects high-level semantics to guide low-level features even when the guide map is partially corrupted. When the occlusion ratio exceeds 30–40%, performance declines more noticeably, underscoring the importance of accurate initial guidance for optimal segmentation.

D. Effect of Input Resolution

To evaluate the effect of input resolution on the performance of the proposed DAD, we conducted experiments with four different input sizes (352, 416, 480, and 512) across multiple COD datasets. As shown in Fig. 12, increasing the input resolution from smaller sizes yields noticeable improvements in segmentation performance. However, beyond a certain point (416, 480, 512), the performance gains become marginal, suggesting that further increasing the resolution offers limited benefit. Therefore, we adopt 480 as the default input size, achieving a good trade-off between accuracy and computational efficiency.

E. Performance Under Multiple Foreground Objects

As illustrated in Fig. 13 and Fig. 14, even with multiple foreground objects present, our prototype extraction effectively captures the essential binary nature of the foreground. This enables robust segmentation by treating all target instances collectively as a single foreground class, demonstrating the adaptability of our method in complex scenarios. Furthermore, the explicit guidance from high-level semantics to low-level

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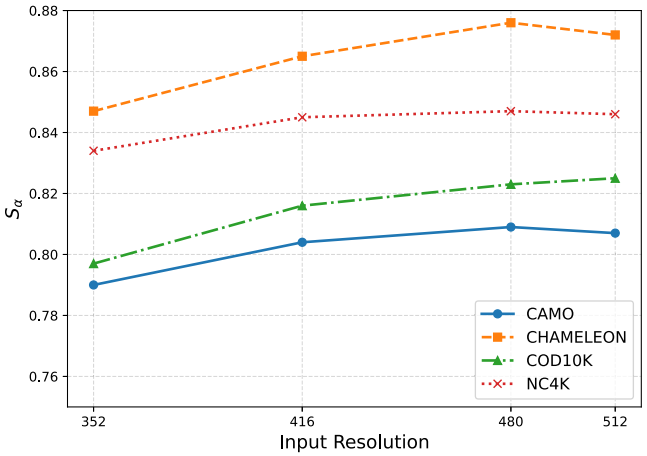


Fig. 12. Effects of different input resolutions across multiple COD datasets.

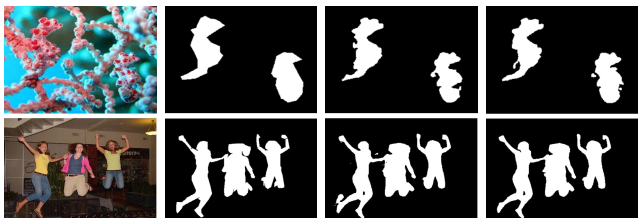


Fig. 13. Visual examples for multiple foreground objects in COD (1st row) and SOD (2nd row) tasks.

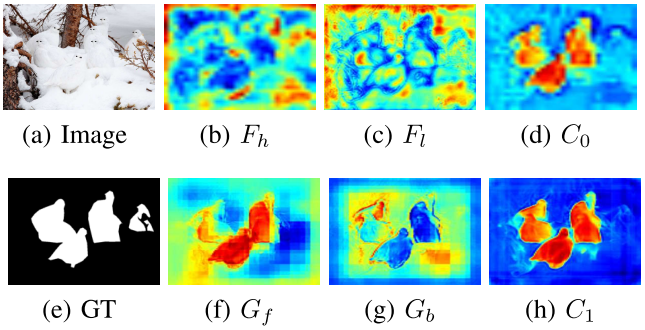


Fig. 14. Feature visualizations at each stage for multiple foreground objects.

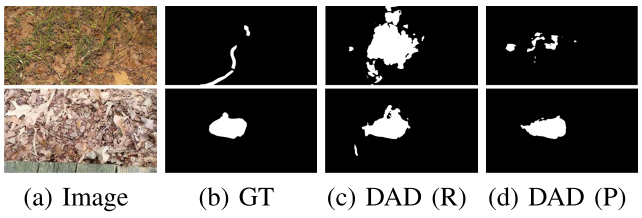


Fig. 15. Visual examples on low-resolution and noisy data.

details enhances the discriminability of the foreground, achieving a highly separable representation.

F. Limitations

As illustrated in Fig. 15, we present failure cases of our method on low-resolution and noisy inputs. It is evident that, due to image blurriness compounded by limited spatial resolution, our method exhibits certain limitations under such conditions. This observation indicates that first performing high-quality image restoration from degraded inputs and then applying segmentation may provide a more effective solution.

In addition, although our framework is efficient in terms of MACs and parameter count, the two-stage design and cross-window attention inevitably introduce additional computational overhead, leading to lower FPS compared with lightweight real-time models. This limitation underscores the importance of further runtime optimization, such as mixed-precision inference or model acceleration techniques, to enhance its practical applicability.

VI. Conclusion

In this paper, inspired by the way human eyes detect objects, we propose a new unified dual-branch decoder paradigm, termed the difference-aware decoder, to better explore the differences between foreground and background and to separate objects of interest in optical images. This decoder operates in two stages, leveraging multi-level features from the encoder. In the first stage, coarse detection of foreground objects is achieved by directly utilizing high-level semantic features, mimicking the initial rough observation of human vision. In the second stage, the decoder refines segmentation by exploring differences in low-level features, guided by the coarse map from the first stage. To enhance this process, we introduce two key innovations: a difference-aware prototype generation method that captures and compares prototype vectors for foreground and background, and an Overlapped Window Cross-Level Semantic Guidance mechanism. This mechanism integrates high-level semantic information into low-level features through channel grouping and multi-scale aligned window pairs, improving the model's ability to discern subtle differences. These innovations enable precise foreground extraction, effectively handling the challenges posed by varying and complex backgrounds. The experiments conducted in this study confirmed the superiority of the proposed difference-aware decoder for SOD and COD. Furthermore, we also proved the effectiveness of the different modules. In the future, we will extend the proposed difference-aware decoder to other binary segmentation tasks, such as road extraction.

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Jiepan Li (Graduate Student Member, IEEE) received the B.S. degree in measurement and control technology from the School of Electronic Information, Wuhan University, Wuhan, China, in 2020. He is currently pursuing the Ph.D. degree in photogrammetry and remote sensing with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University.



Wei He (Senior Member, IEEE) is currently a Full Professor with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University. His research interests include image quality improvement, remote sensing image processing, matrix/tensor analysis, and deep learning.



Fangxiao Lu (Student Member, IEEE) received the B.S. degree from the School of Geomatics and Geodesy, Wuhan University, Wuhan, China, in 2020. He is currently pursuing the Ph.D. degree in photogrammetry and remote sensing with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University.



Hongyan Zhang (Senior Member, IEEE) is currently the Dean of the School of Computer Science, China University of Geosciences, Wuhan. His research interests include image reconstruction for quality improvement, hyperspectral information processing, and agricultural remote sensing.

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